



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment 3 CLO Mapping: CLO-2

Course Code	MAT233	Semester	Fall 2025
Course Title	Linear Algebra		
Instructor	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	12-12-2025	Submission Date	19-12-2025
Total Marks	25 = Assignment (20) + Quiz Based Viva (5)		

Sr.	Name of student	Registration no.	Section	Assignment marks	Viva marks	Total marks
1						

Submission Instructions (Please follow strictly)

- Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty space in the assignment PDF file. DO NOT use any other papers.
- This is an INDIVIDUAL assignment.
- Any question related to assignment topic may be asked in the viva.
- Write your complete names and registration numbers on each page of the assignment.
- Follow the deadline. Finish your work one day before, so you could submit in time.
- LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).
- All Plagiarized assignments will be awarded ZERO marks.

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student signature: _____

Name:

Registration No.

Question No.1

[CLO2]

[10 marks]

Let $V = R^3 = \{(x, y, z); x, y, z \in R\}$ be set of ordered pairs of triples of real numbers with the following operations of addition and scalar multiplication.

$$\vec{u_1} + \vec{u_2} = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 + 2, y_1 + y_2, z_1 + z_2 - 1)$$

$$k(x_1, y_1, z_1) = (kx_1, y_1, -kz_1)$$

- a) Compute $\vec{u_1} + \vec{u_2}$ if $\vec{u_1} = (-1, 2, -3)$ and $\vec{u_2} = (4, -2, 5)$
- b) Compute $k \vec{u_1}$ if $\vec{u_1} = (2, -3, 5)$ and $k = -2$
- c) Find **zero vector** of the set.
- d) Find **additive inverse** of the vector $(3, -2, 4)$
- e) **Prove or disprove** the property $1 * \vec{u_1} = \vec{u_1}$

Solution:

Name:

Registration No.

Name:

Registration No.

Name:

Registration No.

Question No. 2

[CLO2]

[10 marks]

Determine whether V is a vector space or not.

Let V be the set of all 2×2 matrices with real entries and take the vector space operations on V as the following

$$\vec{u} + \vec{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11} + v_{11} & u_{12} + v_{12} - 2 \\ u_{21} + v_{21} + 1 & u_{22} + v_{22} \end{bmatrix}$$

$$k\vec{u} = k \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} k \times u_{11} & 1 \\ 1 & k \times u_{22} \end{bmatrix}$$

Solution:

Name:

Registration No.

Name:

Registration No.
